Equation Sheet: Electrostatics

Intro Physics II • Lectures 6–13 • Knight ch. 22–25

fundamental = know cold (your starting points); **derived** = a line or two from the fundamentals.

Symbols and constants

q, Q	charge (C)	Φ	electric flux ($\mathrm{N{\cdot}m^2/C}$)	U	potential energy (J)
$\vec{E}$	electric field (N/C = V/m)	λ, σ, ρ	line / area / volume charge density	V	potential (V = J/C)
r, d	distance / separation (m)	A	area	k	$8.99 \times 10^9\ \mathrm{N{\cdot}m^2/C^2}$
ε_0	$8.85 \times 10^{-12}\ \mathrm{C^2/(N{\cdot}m^2)}$, $k = 1/(4\pi\varepsilon_0)$		$e = 1.60 \times 10^{-19}$ C		

1. Coulomb's law & the electric field (Knight 22.4–22.5, 23.1–23.2)

fundamental

Force between two point charges (along the line joining them; repulsive for like signs; $F \propto 1/r^2$)
$$F = \frac{k\,|q_1 q_2|}{r^2}, \qquad k = \frac{1}{4\pi\varepsilon_0}$$

Field — force per unit test charge
$$\vec{E} = \frac{\vec{F}_{\text{on } q}}{q}, \qquad \vec{F} = q\vec{E}$$

Field of a point charge (away from +, toward −)
$$E = \frac{k\,|q|}{r^2}$$

Superposition (both are vector sums)
$$\vec{F}_{\text{net}} = \textstyle\sum_i \vec{F}_i, \qquad \vec{E}_{\text{net}} = \sum_i \vec{E}_i$$

2. Continuous charge distributions (Knight 23.3–23.6)

fundamental

Chop into pieces, add up (integrate)
$$\vec{E} = \int d\vec{E}, \qquad dE = \frac{k\,dq}{r^2}$$

Charge densities
$$\lambda = \frac{Q}{L}, \qquad \sigma = \frac{Q}{A}, \qquad \rho = \frac{Q}{V}$$

derived

Infinite line of charge (distance r)
$$E = \frac{\lambda}{2\pi\varepsilon_0\, r} = \frac{2k\lambda}{r}$$

Infinite plane / sheet of charge
$$E = \frac{\sigma}{2\varepsilon_0} \quad \text{(uniform — no } r\text{)}$$

3. Electric flux & Gauss's law (Knight 24.1–24.3)

fundamental

Electric flux through a surface
$$\Phi = \oint \vec{E} \cdot d\vec{A}$$

Gauss's law (any closed surface; only the *enclosed* charge matters)
$$\Phi = \frac{Q_{\text{enc}}}{\varepsilon_0}$$

derived

Uniform field through a flat area
$$\Phi = EA\cos\theta \quad (\theta \text{ from the normal})$$

Charge *outside* a closed surface
$$\Phi_{\text{net}} = 0$$

Sphere / spherical shell, total charge Q ($r > R$)
$$E = \frac{kQ}{r^2} \quad \text{(same as a point charge)}$$

4. Conductors in equilibrium (Knight 24.4–24.6)

fundamental

Field inside the conducting material	$\vec{E} = 0$
Excess charge	all on the outer surface
Field at the outer surface	$\perp$ to the surface, $E = \dfrac{\sigma}{\varepsilon_0}$

derived

Empty cavity inside a conductor	$\vec{E} = 0$ in the cavity (Faraday shielding)
Charge $+q$ placed in the cavity	$-q$ on the inner wall, $+q$ on the outer surface

5. Electric potential energy & potential (Knight 25.1–25.7)

fundamental

Two point charges (keep the signs; $U \to 0$ as $r \to \infty$; one power of r)	$U = \dfrac{k\,q_1 q_2}{r}$
Energy conservation	$K_i + U_i = K_f + U_f\,, \qquad W_{\text{elec}} = -\Delta U$
Potential — energy per unit charge	$V = \dfrac{U}{q}\,, \qquad U = qV \qquad (1\text{ V} = 1\text{ J/C})$
Point charge (scalar; keep the sign; $V \to 0$ as $r \to \infty$)	$V = \dfrac{k\,q}{r}\,, \qquad V_{\text{net}} = \sum_i \dfrac{k\,q_i}{r_i}$

derived

Potential difference from the field	$\Delta V = -\displaystyle\int_i^f \vec{E}\cdot d\vec{s} \xrightarrow{\text{uniform}} \Delta V = -Ed$
Field from the potential (the gradient)	$E_x = -\dfrac{dV}{dx}\,, \qquad \vec{E} = -\nabla V$
Equipotentials	$\perp$ field lines; $\vec{E}$ points toward decreasing V

Companion simulations at `hoppese.github.io/Intro-physics-sims/`: `electroscope`, `e-fields-conductors-and-insulators`, `electrostatics-explorer`, `field-superposition-explorer`, `discrete-charge-builder-3d`, `gauss-flux`, `pe-vs-separation`, `equipotential-explorer`.